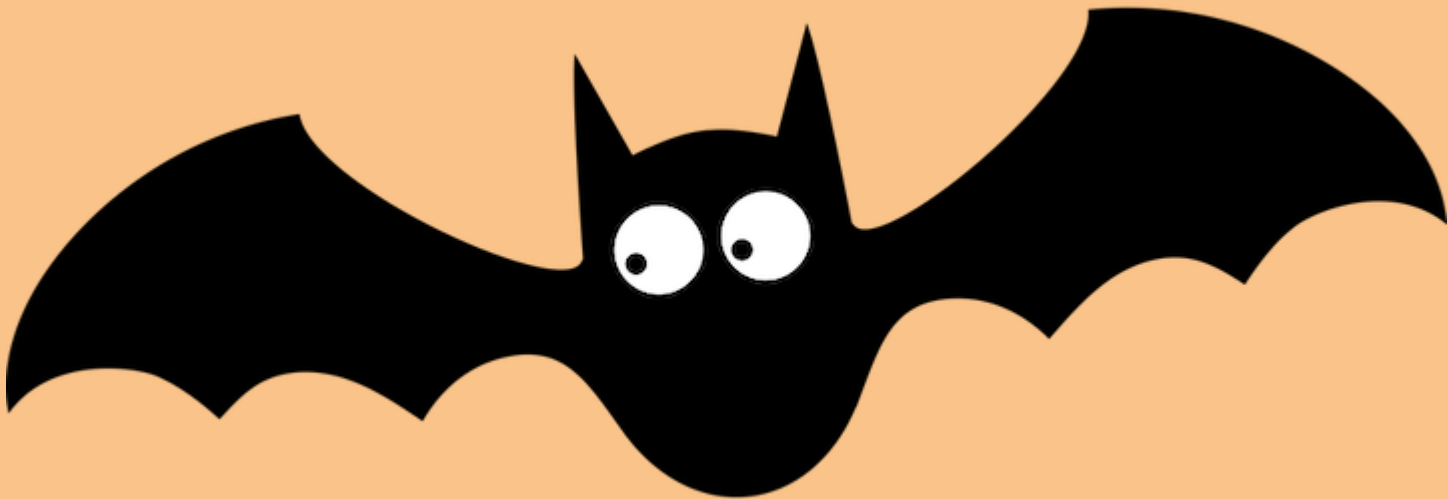




Gradient Boosting

Son Nguyen

(*)

Boosting means "learning from the mistake".

↙
Adaboost

Increase the weights of
the mistakes

↘
Gradient Boosting

replace the original
target (y-value, output
variable) by the errors

(*) Let's predict the temperature tomorrow

Model 1: predict 61°F . This model will make some
error (even when error = 0).

we bring model 2 just to predict the error of model 1

Model 2 will make another error. we bring in model 3 to predict the error of model 2.

And so on.

In the end the final prediction is

$$\text{Predict of Model 1} + \text{Predict of model 2} + \text{Predict of M3} \\ + \dots$$

Let's see an example ;

Data 1:

x	y
60	70
70	75
55	50
67	76
66	59

Data to train model 1 (M1)

λ : Learning rate

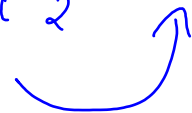


x	y	M1 predicts	error 1
60	70	40	30
70	75	90	-15
55	50	80	-30
67	76	70	6
66	59	100	41



x	error 1
60	30 $\cdot \lambda$
70	-15 $\cdot \lambda$
55	-30 $\cdot \lambda$
67	6 $\cdot \lambda$
66	41 $\cdot \lambda$

Data to train model 2

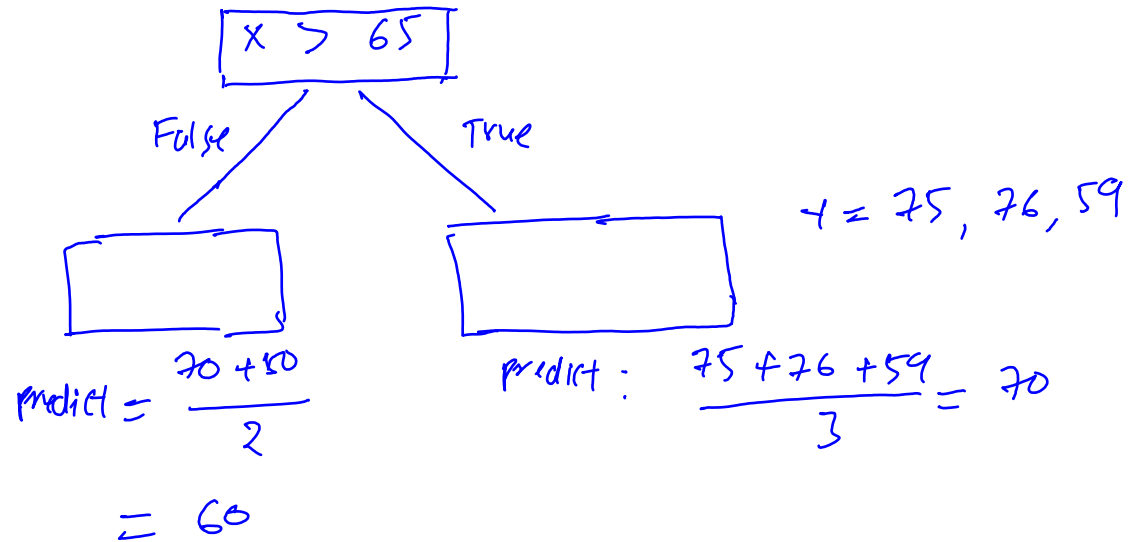


Notice that: Although y is continuous in this case, Gradient Boosting also works when y is categorical. The same goes for AdaBoost.

(*) How decision tree used for continuous y variable.

Example :

X	Y
60	70
70	75
55	50
67	76
66	59



(*) How to evaluate the model when y is continuous
regression problem.

① R^2

② Absolute mean error (AME)

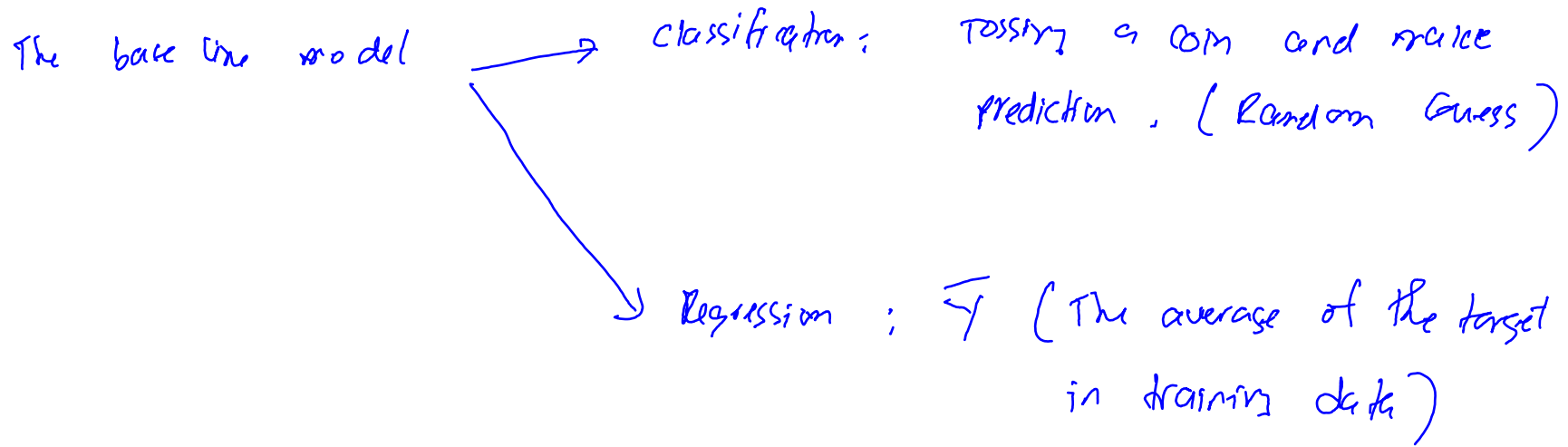
③ MSE : Mean Squared error.

X	y	$\hat{y}$	Absolute error	Squared error
60	70	40	30	900
70	75	90	15	225

$$SSE = \text{Sum squared error} = 900 + 225 = 1125$$

$$MSE = \text{Mean} \frac{1125}{2} = 562.5$$

$$AME = (30 + 15) / 2 = 22.5$$



$$R^2 = 1 - \frac{\text{SSE of the model}}{\text{SSE of the baseline model}}$$

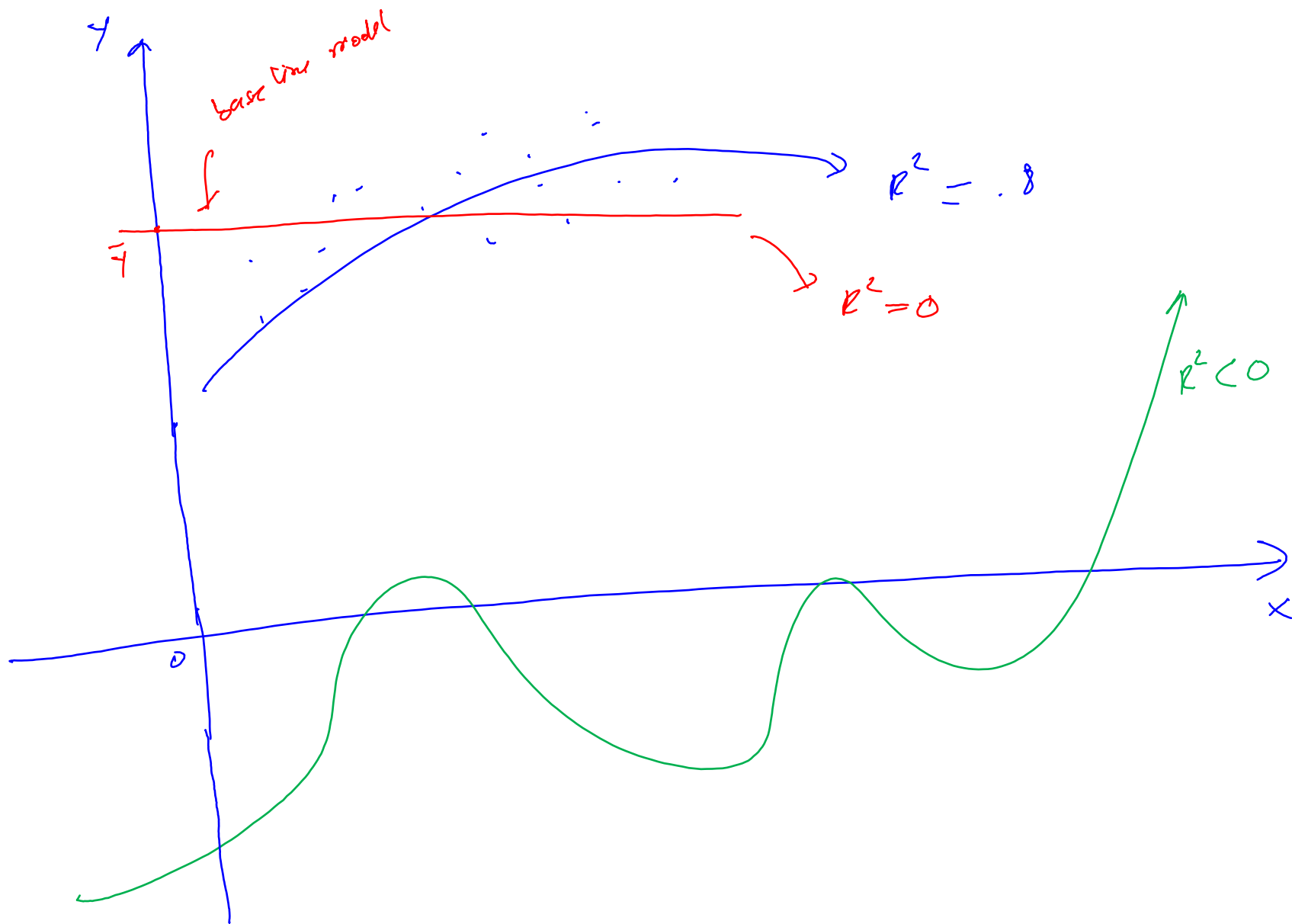
(SSE = Sum Squared Errors)

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$R^2 = 1$: perfect model with no errors

$R^2 = 0$: when the model is baseline model.
($\bar{y}$)

$R^2 < 0$: when the model is worse than $\bar{y}$



(*) Hyper parameters for Adaboost and Gradient Boosting

(1) Number of weak learners

(2) Learning Rate

weak learners are usually Stump or very shallow trees.

